

Austrian theoretical nuclear physicist Lise Meitner coined the term “nuclear fission” and came up with the theory that explained the science behind the splitting of the nuclei of uranium atoms, first carried out by her colleagues in 1938.

MILESTONES

BERLIN RESEARCH

Moves to Berlin in 1907, working as departmental assistant at Max Planck's physics institute.

HEAD OF DEPARTMENT

Awarded Leibniz Medal and made supervisor of the physics section at the University of Berlin in 1917.

SETTLES IN SWEDEN

Escapes Nazi Germany in July 1938, when Austrian citizens become fully subject to German law.

REVOLUTIONARY PAPER

Along with nephew Otto Frisch, publishes theory of nuclear fission in *Nature* on February 11, 1939.

LONG-AWAITED AWARD

With Hahn and Strassman, given the 1966 Enrico Fermi Award by US Atomic Energy Commission.

Lise Meitner grew up in Vienna, the third of eight children in a liberal Jewish family. Her father Philipp was a lawyer who, together with his wife Hedwig, fostered an intellectually stimulating environment for their children. Meitner and her siblings were often present among regular gatherings of writers, lawyers, legislators, politicians, and chess players in the family home.

At the age of 8, Meitner kept a mathematics notebook under her pillow and asked probing questions of the world around her, including ones pertaining to reflected light and the patterned sheen on oil spills. Recognizing her intellect, and believing his daughters should enjoy the same education as his sons, Philipp ensured Meitner was privately tutored after she turned 14, the age at which girls in Austria were barred from state schooling.

Female trailblazer

The objective of Meitner's private tuition was to pass the entrance exam for the University of Vienna. She passed in July 1901, and later that year, aged 23, became one of the first women to attend the university's physics course. Having been tutored by the brilliant theoretical physicist Ludwig Boltzmann, in February 1906, Meitner became only the second woman at the university to receive a physics doctorate. Her thesis was on how heat travels through inhomogeneous solids (in which particles are not evenly distributed) and proved that they conduct heat in a similar way to electricity. This was in

Having fled to Sweden, Meitner was initially unable to see the results of her nuclear fission experiment. She explained the process of fission using data that Hahn sent by letter.

